

Effects of Observing a Robot Expressing Approach-Avoidance Conflict on Observers' Self-Evaluations of Personal Courage

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2 ABSTRACT

3 This study examined whether a robot's externalization of approach-avoidance conflict before
4 action influences observers' self-evaluations of personal courage. Personal courage involves not only
5 moving toward a valued action but also the internal state of experiencing fear or hesitation before
6 acting. In this study, speech bubbles were displayed near the head of a cylindrical robot using
7 projection mapping, and approach and avoidance motives were presented in a scene in which the
8 robot confronted littering in a park.

9 Study 1 compared conflict and no-conflict conditions, as well as sequential and simultaneous
10 presentation methods, to examine whether a robot expressing approach-avoidance conflict was
11 perceived as courageous. Study 2 used the simultaneous presentation selected in Study 1
12 and examined how the presence or absence of conflict and admonishing behavior influenced
13 post-stimulus self-evaluations of personal courage in relation to observers' pre-existing courage
14 tendency.

15 In Study 1, the conflict condition produced higher courage ratings and conflict ratings than the no-
16 conflict condition, confirming that the presented internal state was perceived as conflict. In Study
17 2, the interaction between pre-existing courage tendency and conflict was significant: participants
18 in the low-courage group tended to show higher self-evaluations in the conflict condition, whereas
19 participants in the high-courage group showed lower self-evaluations in the conflict condition. In
20 contrast, the presence or absence of admonishing behavior showed no clear effect. These findings
21 suggest that visually externalizing approach-avoidance conflict through a robot may influence
22 observers' self-evaluations of personal courage. However, this influence may not be uniform and
23 may differ depending on observers' pre-existing courage tendency.

24 Keywords: personal courage, approach-avoidance conflict, observational learning, human-robot interaction, internal state, robot
25 expression, social modeling

1 INTRODUCTION

26 In everyday life, people sometimes need to move toward valued actions even in situations that
27 involve fear or anxiety. In this study, we refer to the capacity to move toward such actions as

personal courage. Courage has been conceptualized not as the absence of fear but as moving toward meaningful action in the presence of fear (Rachman, 1984; Woodard and Pury, 2007). Personal courage is important because it allows the difficulty of an action to be understood in relation to the actor's own context. Pury et al. distinguished between general courage, which is evaluated as courageous by most people, and personal courage, which is courageous in light of a person's life context and personal constraints (Pury et al., 2007). From this perspective, a child with a learning disability going to school on the day of a major test, or a person wrapping Christmas presents for the first time after being unable to do so for many years because of a history of PTSD, can be understood as actions involving high fear and difficulty in that person's context (Pury et al., 2007). Furthermore, actions such as speaking up, seeking help, or pointing out a problem may be valuable for the person or others, but they also involve social and psychological risks such as lowered evaluation and rejection. These actions are therefore meaningful targets for investigation from the perspective of personal courage (Howard et al., 2017; Vogel et al., 2007; Milliken et al., 2003).

However, people do not always act even when they recognize that an action is valuable. For example, expressing an opinion in front of others often involves anxiety about how one will be evaluated (Watson and Friend, 1969; Dickerson and Kemeny, 2004). Consulting others or seeking help can also involve concern about being viewed negatively (Vogel et al., 2007). In workplaces, many employees report having noticed problems or concerns but not communicated them to supervisors, and the main reasons include fear of being viewed negatively or damaging important relationships (Milliken et al., 2003). In other words, speaking up and seeking help can be beneficial for oneself and others, yet they involve social and psychological risks such as lowered evaluation, damaged relationships, reduced status, and loss of self-image. Therefore, supporting personal courage requires more than helping people understand that an action is valuable; it also requires supporting the process of moving toward action while perceiving risk.

As one way to support this process, the present study focuses on observational learning. Observational learning refers to learning in which observing another person's behavior or success increases one's expectation that "I may be able to do it too" (Bandura, 1977). The effectiveness of observational learning was first demonstrated in learning actions that are visible from the outside. For example, children who were afraid of swimming showed better skill outcomes after observing same-age models swimming (Weiss et al., 1998). In studies of sit-to-stand and stand-to-sit action observation, observers' sensorimotor systems respond, and action observation itself can activate motor representations (Chaisaen et al., 2020). At the same time, observational learning can also involve psychological aspects such as effort and persistence, not only externally visible movements. Schunk and Hanson showed that observing peer models affects children's self-efficacy and achievement behavior (Schunk and Hanson, 1985). In addition, infants who observed an adult struggling to achieve a goal made more attempts on a subsequent difficult task than infants who observed an adult easily succeeding (Leonard et al., 2017). These findings suggest that observing others facing difficulty can support observers' sense that they too can try and can encourage persistence.

The purpose of this study is to clarify how presenting personal courage as an object of observational learning affects observers' self-evaluations of their own personal courage. In personal courage, not only "what was done" but also "how frightening it was" and "what kind of hesitation was involved" are important (Pury et al., 2007). However, when a human model is used, it is difficult

to present fear or conflict before action in the same form and intensity across conditions. Therefore, in this study, we used a robot to express a state in which the motivation to move toward a valued action and the motivation to avoid risk or disadvantage coexist as approach-avoidance conflict, and we examined whether observing this state influences observers' self-evaluations of personal courage.

We conducted two studies. Study 1 clarified what kind of robot expression is perceived as a "courage-evoking action." Specifically, we examined whether a robot expressing approach-avoidance conflict is perceived as more courageous than a robot not expressing conflict, and we assessed the validity of the stimulus to be used in Study 2. Study 2 examined whether observing the robot that was confirmed in Study 1 to be perceived as courageous influences observers' self-evaluations of personal courage. In addition, to separate the effect of presenting approach-avoidance conflict from the observational effect of seeing a valued action, we independently manipulated the presence or absence of conflict and the presence or absence of action. Through this design, the present study provides insight into the meaning of presenting not only action itself but also the internal state before action when personal courage is treated as a target of observational learning. It also provides methodological implications for using robots to visualize internal states that are difficult to observe from the outside and to support observational learning related to courage.

2 RELATED WORK

2.1 Personal Courage and Internal States Before Action

Research on courage has treated courage not as the absence of fear but as moving toward action while experiencing fear. Rachman argued that even people with strong fear can perform courageous acts (Rachman, 1984). Norton and Weiss examined the relationship between behavioral approach and courage in a fear-eliciting situation (Norton and Weiss, 2009). Woodard and Pury also conceptualized courage as a construct involving action for a meaningful purpose in a situation involving threat or fear (Woodard and Pury, 2007). These studies provide a basis for understanding courage not as mere risk preference or low fear but as approach behavior in the presence of fear.

Among these studies, Pury et al. distinguished between general courage, which is evaluated as courageous by most people, and personal courage, which is courageous in light of a person's life context and personal constraints (Pury et al., 2007). Highly personal courageous actions are characterized by fear, difficulty, and personal constraints. From this distinction, the same action may or may not be courageous depending on what kinds of fear or difficulty the actor experienced. Therefore, research on personal courage needs to consider not only externally visible behavior but also the internal states preceding that behavior.

Recent models explaining the mechanisms by which courage emerges also focus on these internal states before action. Chowkase et al. describe courage as a sequence involving situation perception, evaluation of value or meaning, evaluation of action feasibility, and action decision (Chowkase et al., 2024). In this process, reasons to move toward a valued action and reasons to avoid risks or disadvantages associated with the action may coexist. This can be understood as approach-avoidance conflict, in which approach and avoidance motives operate simultaneously (Lewin, 1931; Miller, 1944). However, these studies clarify the definition and process of courage; they do not directly examine whether observing another agent's internal state changes observers' self-evaluations of personal courage.

2.2 Observational Learning and Dependence on Observers' Prior Characteristics

Research on observational learning has shown that observing others' behavior and its outcomes can change observers' behavior, motivation, and self-efficacy. Bandura positioned vicarious experience as one source of self-efficacy and argued that observing another person succeed at a task can influence the expectation that "I can do it too" (Bandura, 1977). Schunk and Hanson showed that observing peer models affects children's self-efficacy and achievement behavior (Schunk and Hanson, 1985). These studies suggest that observing others act can influence the cognition that observers themselves can also perform the action.

The effect of a model can also differ depending on the state in which the model acts. Schunk et al. compared mastery models who succeeded easily from the beginning with coping models who showed difficulty and gradually coped with it, and found that observing coping models increased self-efficacy and task performance (Schunk et al., 1987). Schunk and Hanson also showed that models who cope while showing difficulty and negative affect influence learning self-efficacy (Schunk and Hanson, 1989). Leonard et al. showed that infants who observed an adult persisting in an attempt to achieve a goal made more attempts on a subsequent difficult task than infants who observed an adult easily succeed (Leonard et al., 2017). However, the relative advantage of coping models over mastery models has not been consistently observed. Schunk and Hanson found that children who observed a peer model showed higher self-efficacy and achievement than those who observed a teacher model or no model, but reported no significant differences between the mastery- and coping-model conditions (Schunk and Hanson, 1985). Taken together, these findings suggest that observing a process involving difficulty and effort can support observers' self-efficacy and persistence, although coping models do not necessarily outperform mastery models.

At the same time, the same model may be received differently depending on observers' prior characteristics. Braaksma et al. showed that similarity between model and observer and the model's level of expertise are important in observational learning (Braaksma et al., 2002). Lucas et al. showed that people tend to rely on others' judgments and behavior as cues when a task is difficult, whereas people with high self-efficacy are less susceptible to such social influence (Lucas et al., 2006). These findings indicate that, when using observational learning, it is necessary to consider not only the model's characteristics but also observers' prior characteristics, which influence how the model is received. Therefore, the present study also considers observers' prior characteristics when examining the effect of a robot's conflict expression.

2.3 Modeling by Robots and Artificial Agents

Models in observational learning are not limited to humans. Studies using artificial agents and avatars have shown that visually present agents and avatars similar to the self can influence observers' attitudes, self-efficacy, and subsequent behavior (Rosenberg-Kima et al., 2008; Fox and Bailenson, 2009). Research on social robots has also shown that people can treat robots as social models and engage in modeling based on their behavior and social roles (Xu, 2023). It has also been reported that the way a robot encourages participants influences the way participants subsequently encourage others (Higashino et al., 2023).

These studies indicate that robots and artificial agents can function as social models that influence observers. However, the studies reviewed here targeted attitudes, self-efficacy, movement, and encouragement behavior, not personal courage itself. To the extent reviewed in this manuscript,

few studies have explicitly presented what kind of fear or conflict a model has before action and examined how observing it influences observers' self-evaluations of personal courage. Therefore, using a robot to present internal states related to personal courage in an observable form extends existing modeling research.

2.4 Externalizing Internal States Through Robots

The experimental reason for using a robot in this study is that visual expressions, in addition to behavior and speech, can be more easily controlled across conditions. With a human model, observers can infer internal states from facial expressions, voice, silence, or self-disclosure. However, to present the presence or absence of fear, hesitation, approach motives, and avoidance motives in the same scenario and in a comparable form across conditions, an artificial medium is more suitable. Therefore, in this study, we used speech-bubble expressions in addition to the robot's movement and speech, and presented the robot as a stimulus in which internal states were externalized.

Regarding speech-bubble expressions, Nitada et al. proposed a method in which a speech bubble is displayed near a robot's head in a video-conferencing situation, with the user's face drawn inside the bubble (Nitada et al., 2021). They found that the speech-bubble condition increased the sense that the robot was looking at the user. This study suggests that speech bubbles can function as a means of visually representing the robot's attention or mental focus. However, that study addressed the direction of attention, not the conflict between approach and avoidance motives related to courage.

Taken together, prior studies have shown that personal courage involves actions accompanied by fear or difficulty, that observational learning can influence self-efficacy and perceived action possibility, and that robots and artificial agents can function as social models. However, to the extent reviewed in this manuscript, research has not sufficiently examined whether externalizing, through a robot, internal states before action related to personal courage, especially conflict between approach and avoidance motives, affects observers' self-evaluations of personal courage. The present study focuses on this unexplored issue and examines how observing a robot expressing approach-avoidance conflict relates to observers' self-evaluations of personal courage.

3 DESIGN OF THE VIDEO STIMULI

3.1 Robot and Projection Environment

The robot used in this study had the same body as the switch-based robot interface proposed by Omichi et al. (ICSR 2025). This robot has a cylindrical body, and an agent with a face emerges from inside the body and speaks (Omichi et al., 2026).

This study examined whether presenting the robot's internal state before action through speech bubbles influences observers' self-evaluations of personal courage. Therefore, it was necessary to minimize interpretive variability arising from the robot's appearance or bodily movements. Unlike humanoid robots, this robot does not involve complex facial expressions or limb movements; its main expressions are limited to face presentation and speech. Therefore, we considered it suitable as a stimulus medium for presenting approach motives, avoidance motives, and the presence or absence of admonishing speech in a relatively controlled manner. For the admonishing speech, prerecorded audio was used and played by the experimenter through keyboard operation.

192 In the video stimuli, the robot was filmed so that it appeared in the lower-right corner of the
193 frame, and the robot's internal state was displayed as a speech bubble projected by a projector onto
194 the wall behind the robot. The speech bubble was positioned so that it appeared near the robot's
195 head and was used not as the robot's speech but as a visual expression of the internal state before
196 action.

197 3.2 Stimulus Scenario

198 In this study, we used video stimuli depicting a scene in which a robot sees a person littering in
199 a park. Littering in a park is easily understood as a violation of shared norms in a public space.
200 Indeed, littering has been used as a representative task for examining the effects of descriptive and
201 injunctive norms (Cialdini et al., 1990). Observational research in public spaces has also shown
202 that the presence of existing litter increases littering, whereas the availability of trash bins reduces
203 it. Thus, littering is positioned as an everyday norm violation that is affected by environmental
204 and normative cues (Schultz et al., 2013). In addition, cues of disorder such as litter and graffiti
205 can induce other norm violations (Keizer et al., 2008). Therefore, admonishing the person in this
206 scene has value in terms of keeping the park clean, creating an opportunity for the person to stop
207 littering, and reaffirming the norm that littering is unacceptable. At the same time, admonishing a
208 stranger involves social and psychological risks such as being yelled at, ignored, or viewed negatively.
209 Littering in a park has also been used as a stimulus scenario in prior research on social control
210 behavior and has some validity as a scene involving intervention in a stranger's norm violation
211 (Chekroun and Brauer, 2002). Field research has further shown that sanctions against strangers'
212 norm violations include not only direct admonition but also indirect methods, and that direct
213 admonition can involve costs such as the possibility of retaliation (Balafoutas et al., 2014). Research
214 on social courage has identified moving toward valued action while facing interpersonal risk as an
215 important feature (Howard et al., 2017). Therefore, admonishing littering in a park is suitable
216 as a stimulus scenario for examining personal courage because it involves both valued action and
217 interpersonal risk, and it can elicit hesitation or conflict for the actor, even though it is not a
218 situation involving extremely high physical danger.

219 An example of the scenario description in the stimulus is shown in Figure 1.

220 3.3 Common Structure of the Video Stimuli

221 The video stimuli did not present the park scene itself as live-action footage; instead, the scene was
222 constructed using a textual scenario description and speech-bubble expressions. At the beginning
223 of the video, the following scenario description was presented: "The robot saw a person littering
224 in a park." The robot then showed situation recognition in the speech bubble, such as "Oh, there
225 is a person littering," followed by cognition indicating that the situation required admonition.
226 The subsequent presentation of motives and the presence or absence of admonishing speech were
227 manipulated according to the conditions in each study. All videos were created at a resolution of
228 1280 x 720 pixels. The videos in Study 1 were approximately 50–54 seconds long, and the videos
229 in Study 2 were approximately 49–50 seconds long.

230 3.4 Approach Motives and Avoidance Motives

231 In this study, reasons for moving toward a valued action were defined as approach motives, and
232 reasons for wanting to avoid the risks or disadvantages associated with the action were defined as

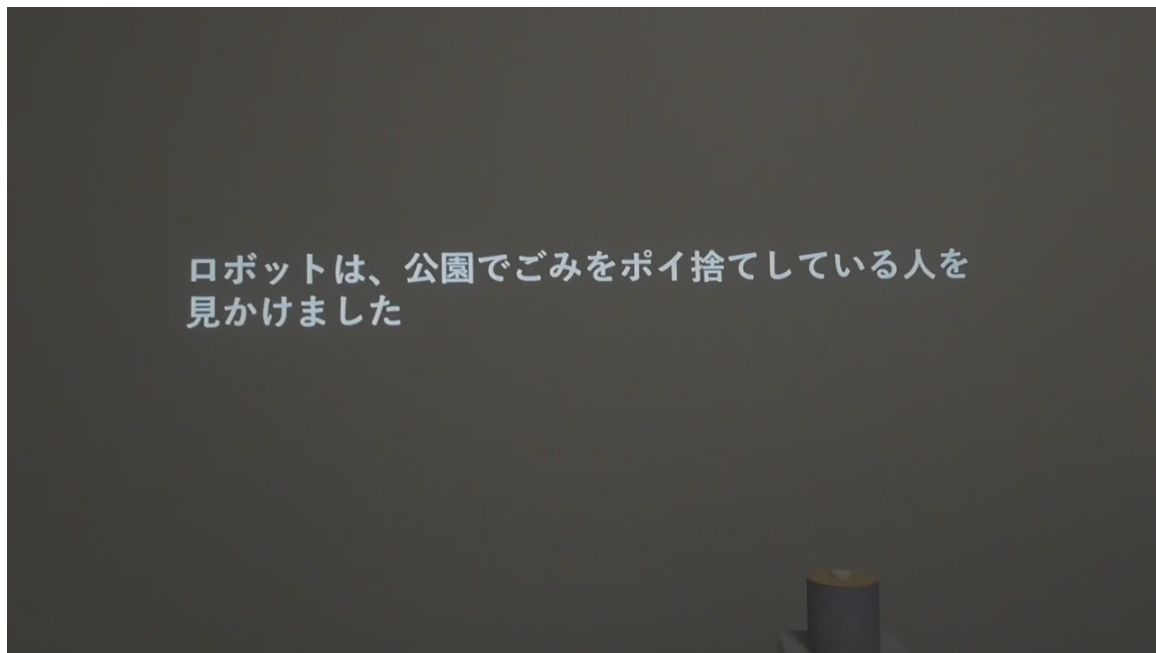


Figure 1. Example of the scenario description in the video stimulus. Text presentation of the situation in which the robot sees a person littering in a park.

avoidance motives. The specific motive statements are shown in Table 1. The approach motives indicated the possibility that desirable outcomes would result from admonishing behavior, whereas the avoidance motives indicated social and psychological risks that could result from admonishing behavior. By combining these approach and avoidance motives, the study expressed a state in which reasons for moving toward a valued action and reasons for avoiding that action coexist. This state is positioned as approach-avoidance conflict, in which approach and avoidance motives operate simultaneously (Lewin, 1931; Miller, 1944).

Table 1. Motive statements used in the video stimuli.

Type of motive	Motive statement	Content
Approach motive	"If I warn them, the park might become cleaner."	Possibility that admonishing behavior improves the public space
Approach motive	"If I warn them, it might prompt that person to stop littering."	Possibility that admonishing behavior changes the other person's behavior
Avoidance motive	"I would hate it if they yelled at me after I warned them."	Risk of receiving an aggressive response from the other person because of admonishing behavior
Avoidance motive	"Even if I warn them, they might ignore me and not take me seriously."	Risk that the admonition is not accepted and is ignored

An example in which the approach and avoidance motives were presented simultaneously is shown in Figure 2.

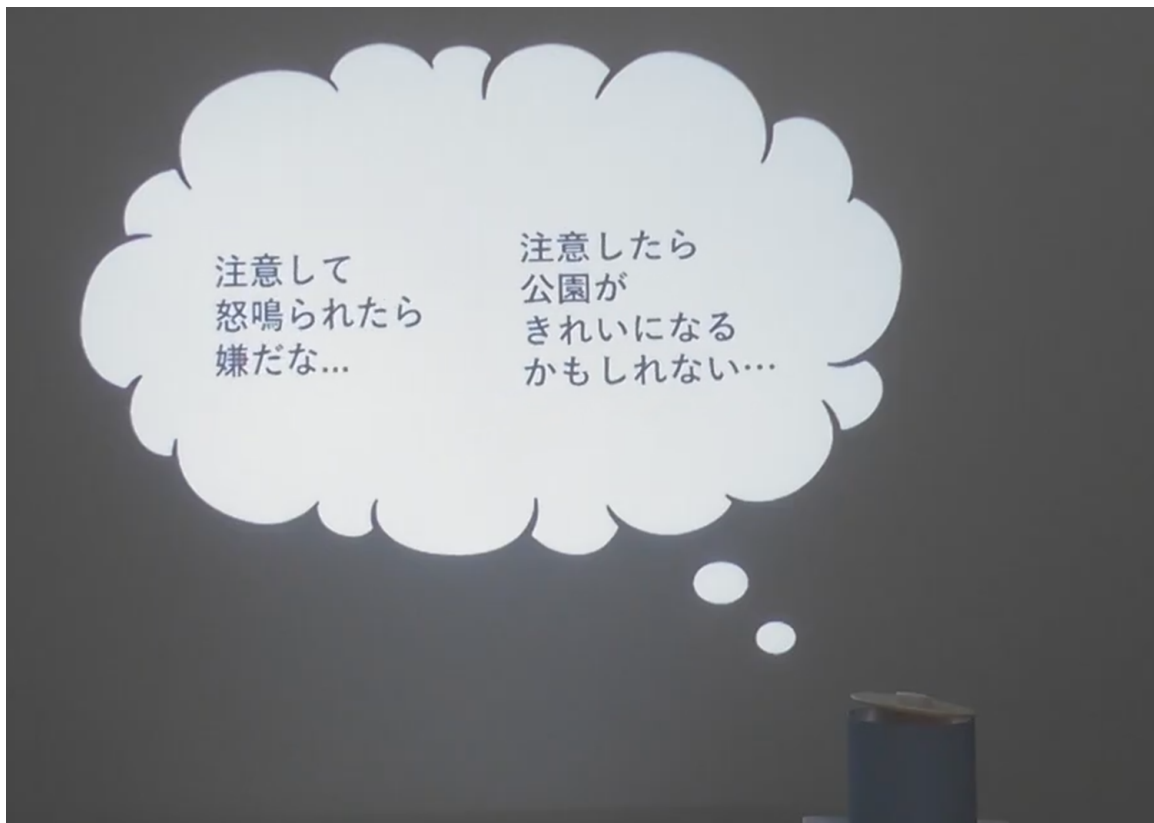


Figure 2. Example of simultaneous presentation of approach and avoidance motives. Simultaneous display of a desirable consequence of admonishing and a concern about admonishing.

242 Through this common design, the present study presented internal states before action in a form
 243 that observers could visually understand. The details of the condition manipulations in each study
 244 are described in the respective method sections.

4 STUDY 1: IS A ROBOT EXPRESSING APPROACH-AVOIDANCE CONFLICT PERCEIVED AS COURAGEOUS?

245 4.1 Purpose

246 The purpose of Study 1 was to examine whether the presentation to be used in Study 2 would be
 247 perceived as courageous. Specifically, we examined whether a robot expressing approach-avoidance
 248 conflict as an internal state would be perceived as more courageous than a robot not expressing
 249 conflict. We also used conflict ratings to confirm whether the presented internal state was perceived
 250 as approach-avoidance conflict and explored which presentation method should be selected for
 251 Study 2.

252 Approach-avoidance conflict is described as a state in which approach and avoidance motives
 253 operate simultaneously (Lewin, 1931; Miller, 1944). At the same time, the experience of conflict
 254 may also occur as temporal fluctuation. Therefore, in Study 1, we considered two candidate methods:
 255 sequential presentation, in which approach and avoidance motives were presented in sequence, and
 256 simultaneous presentation, in which both motives were presented at the same time. In Study 1,

courage ratings were prioritized, and if no difference was found, conflict ratings were used as a reference for selecting the presentation method for Study 2.

Based on this reasoning, we predicted that the conflict condition would produce higher robot courage ratings than the no-conflict condition. As a manipulation check, we also predicted that the conflict condition would produce higher conflict ratings than the no-conflict condition. The comparison between sequential and simultaneous presentation was exploratory, with the aim of examining which method would be perceived as more courageous.

4.2 Method

4.2.1 Conditions

Study 1 used a within-participant design with conflict and presentation method as factors. Conflict had two levels: no conflict and conflict. Presentation method had two levels: sequential presentation and simultaneous presentation. To avoid confounding the effect of the presence or absence of final action, all conditions in Study 1 presented the robot performing the admonishing action.

In the stimulus videos, after the robot saw a person littering in a park, it showed situation recognition and motives through a speech bubble. In the conflict condition, approach and avoidance motives were presented; in the no-conflict condition, only approach motives were presented. In the simultaneous presentation condition, approach and avoidance motives were presented simultaneously and alternately enlarged and reduced. In the sequential presentation condition, the two motives were presented in sequence. Finally, in all conditions, the robot said, “Excuse me, this is not a place to throw away trash.”

Representative frames related to the presentation methods in the conflict conditions of Study 1 are shown in Figure 3. In Figure 3, the scenario description, situation recognition, and cognition indicating that the situation required admonition are omitted; only the portions presenting the approach and avoidance motives are shown. In the sequential presentation condition, the avoidance motive “I would hate it if they yelled at me after I warned them” and the approach motive “If I warn them, the park might become cleaner” were presented in separate frames in sequence. In the simultaneous presentation condition, the same avoidance and approach motives were presented simultaneously within a single speech bubble and alternately emphasized.

4.2.2 Measurement

The dependent variables were robot courage ratings and conflict ratings. Each rating was obtained on a 7-point scale after participants observed each video, and the mean of the items was used as the score. Courage ratings were created by modifying the six items of the Japanese version of the Courage Measure (CM-J) for rating the robot (Shimotsukasa et al., 2023). The CM-J is a scale that measures the tendency to act despite fear or anxiety, and its reliability and validity have been confirmed (Shimotsukasa et al., 2023). In Study 1, to measure the extent to which the robot appeared courageous rather than the observer’s own courage, the subject of each item was changed to “This robot,” and item endings were changed to observational rating expressions such as “appeared to” and “was.” In the analysis, the mean of the six items was used as the courage rating score.

Conflict ratings were measured using original items to assess the extent to which the robot appeared to waver between approach and avoidance motives. In creating the items, based on

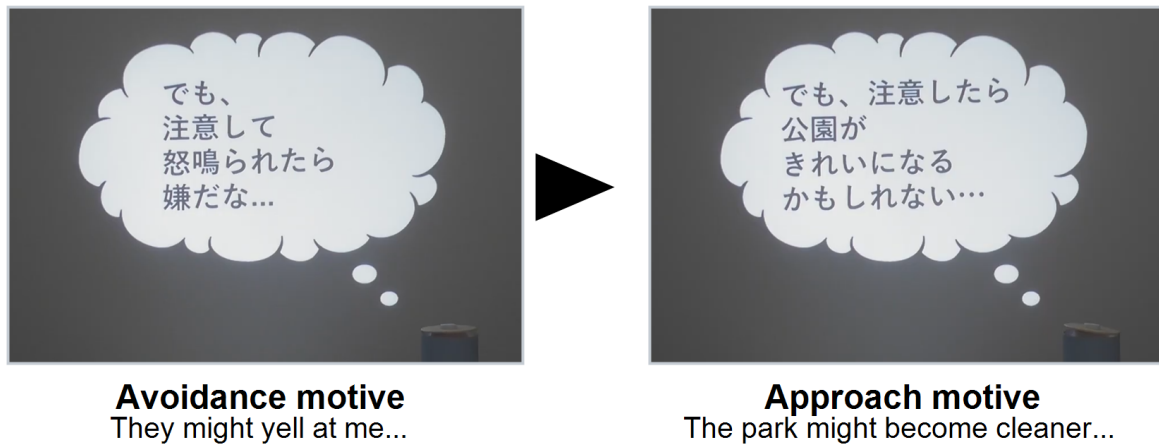
Table 2. Study 1 conditions.

Study 1 condition	Internal-state content	Motive statements used	Presentation method	Final admonishing speech
No conflict, sequential presentation	Approach motives only	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering”	Approach motives presented in sequence	Present
No conflict, simultaneous presentation	Approach motives only	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering”	Multiple approach motives presented simultaneously and alternately emphasized	Present
Conflict, sequential presentation	Approach and avoidance motives	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering” / Avoidance motives: “I would hate it if they yelled at me after I warned them”; “Even if I warn them, they might ignore me and not take me seriously”	Approach and avoidance motives presented in sequence	Present
Conflict, simultaneous presentation	Approach and avoidance motives	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering” / Avoidance motives: “I would hate it if they yelled at me after I warned them”; “Even if I warn them, they might ignore me and not take me seriously”	Approach and avoidance motives presented simultaneously and alternately emphasized	Present

the definition of approach-avoidance conflict (Lewin, 1931; Miller, 1944), we verbalized observable aspects of approach-avoidance conflict: wanting to do the thing in front of oneself but hesitating because of fear, having hesitation or fluctuation in action, having mixed feelings of wanting and not

Study 1: Conflict Presentation

(A) Sequential: motives shown in separate frames



(B) Simultaneous: both motives shown in one frame

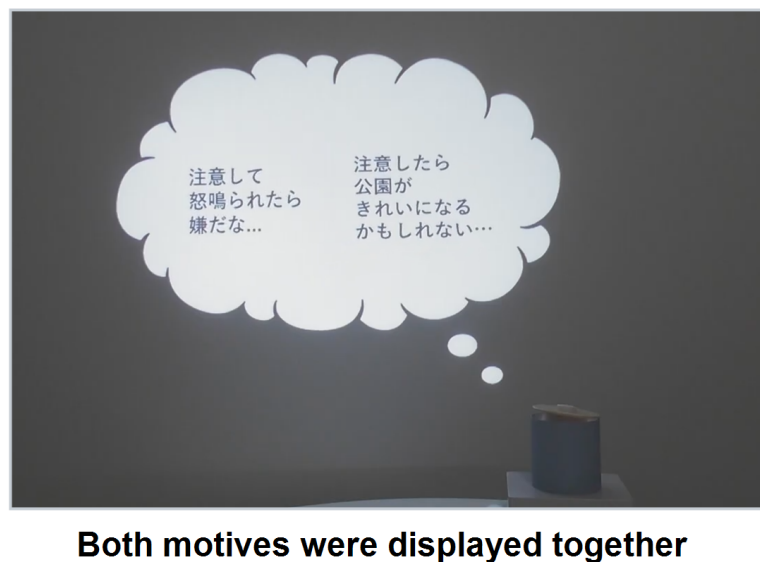


Figure 3. Comparison of conflict presentation methods in Study 1. (A) In the sequential presentation condition, the avoidance and approach motives were shown in separate frames. (B) In the simultaneous presentation condition, both motives were shown within a single frame.

Table 3. Courage-rating items.

Item	Courage-rating item
1	This robot appeared to confront its own fear.
2	This robot appeared not to run away until it did what it had to do, even if it felt strong fear.
3	This robot did something even though it seemed dangerous.
4	This robot took action or confronted the situation anyway, even though it had some worry or anxiety.
5	This robot confronted something frightening when there was an important reason to confront it.
6	This robot appeared not to back down, even when something threatened it.

301 wanting to act, and being unable to decide one's action. In the analysis, the mean of the following
 302 four items was used as the conflict rating score. In addition, to confirm whether ratings based on
 303 these four items corresponded to a direct judgment of conflict, we also obtained a direct item: "This
 304 robot appeared to be experiencing conflict."

Table 4. Conflict-rating items.

Item	Conflict-rating item
1	This robot appeared to want to do what was in front of it but to hesitate because of fear.
2	This robot appeared to have hesitation or fluctuation in its own action.
3	This robot appeared to have mixed feelings of wanting and not wanting to act.
4	This robot appeared unable to decide its own action.

305 4.2.3 Procedure

306 Study 1 was conducted on Yahoo! Crowdsourcing. Participants viewed videos from the four
 307 conditions and, after each video, responded to the robot courage ratings and conflict ratings. The
 308 order of video presentation was randomized for each participant. The response data included video-
 309 content comprehension checks and attention-check items.

310 4.2.4 Participants

311 Exclusions were made based on video-content comprehension checks and attention-check items,
 312 resulting in a final analytic sample of 131 participants. Participants were 18 to 29 years old, with a
 313 mean age of 24.26 years ($SD = 3.36$). The sample included 76 women, 52 men, and 3 participants
 314 who selected "do not know/prefer not to answer."

315 4.2.5 Statistical Analysis

316 In Study 1, robot courage rating scores and conflict rating scores were treated as dependent
 317 variables, and two-way repeated-measures analyses of variance were conducted with conflict (no
 318 conflict vs. conflict) and presentation method (sequential vs. simultaneous) as within-participant
 319 factors. Generalized eta squared was reported as the effect size. For the courage ratings and conflict
 320 ratings, internal consistency was examined using Cronbach's alpha, and the validity of treating the
 321 mean of each item set as a scale score was supplementarily confirmed by factor analysis specifying
 322 a one-factor solution. When an interaction was significant, paired comparisons between conditions
 323 were examined using Wilcoxon signed-rank tests. Analyses were conducted using Python packages
 324 pandas, pingouin, and SciPy.

325 4.3 Results

326 First, we examined whether the courage ratings and conflict ratings could each be treated as a
 327 single scale score. Because the sign of factor loadings obtained in factor analysis is arbitrary, factor
 328 loadings are reported as absolute values. Cronbach's alpha for the six courage-rating items was
 329 .925. In the factor analysis specifying a one-factor solution, the factor loadings on the first factor
 330 ranged from .727 to .855, and the contribution rate was 68.0%. The first eigenvalue was 4.396,
 331 and all subsequent eigenvalues were below 1. Therefore, the mean of the six items was used as the
 332 courage rating score. Cronbach's alpha for the four conflict-rating items was .928. In the factor
 333 analysis specifying a one-factor solution, the factor loadings on the first factor ranged from .845

to .898, and the contribution rate was 76.5%. The first eigenvalue was 3.293, and all subsequent eigenvalues were below 1. Furthermore, a high positive correlation was found between the factor score calculated from the four items and the direct item ($r = .887$, $p < .001$). Therefore, the mean of the four items was used as the conflict rating score.

For courage ratings, the conflict condition was rated higher than the no-conflict condition. The main effect of conflict was significant ($F(1, 130) = 12.216$, $p = .000649$, generalized eta squared = .020). In contrast, the main effect of presentation method was not significant ($F(1, 130) = 2.657$, $p = .106$), and the interaction was also not significant ($F(1, 130) = 0.906$, $p = .343$). Condition means are shown in Figure 4.

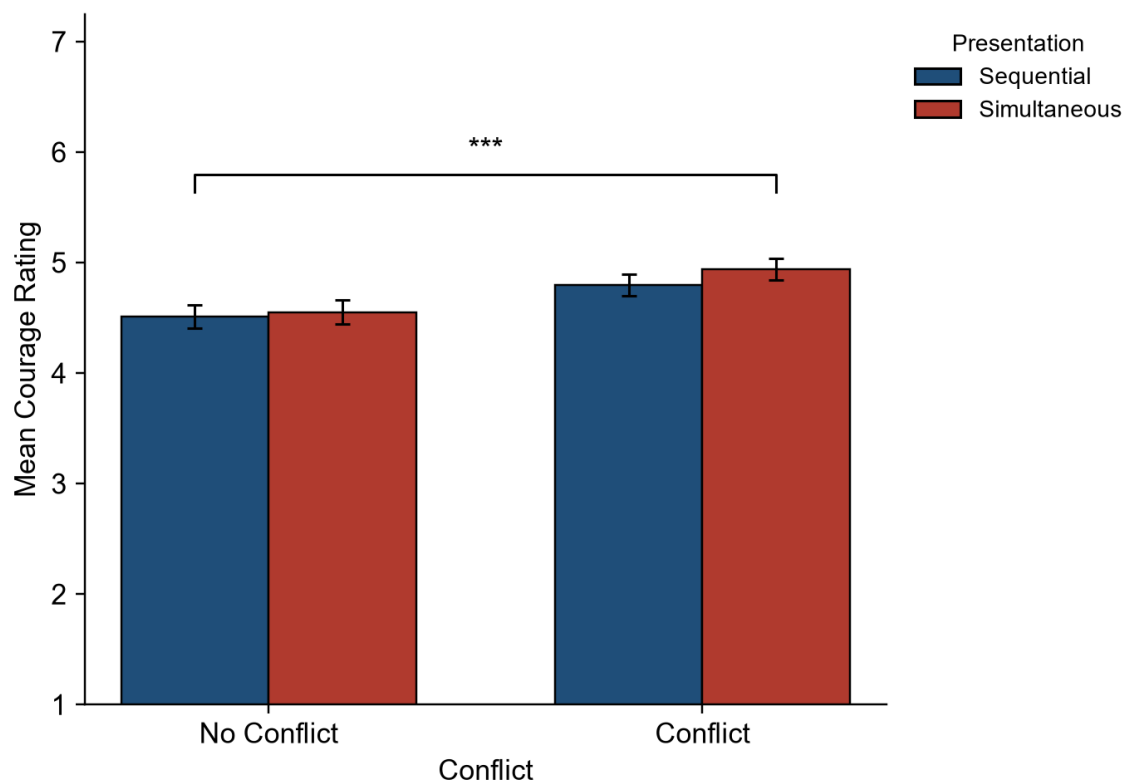


Figure 4. Condition means of courage ratings in Study 1.

For conflict ratings, the conflict condition was rated higher than the no-conflict condition, and simultaneous presentation was rated higher than sequential presentation. The main effect of conflict was significant ($F(1, 130) = 79.558$, $p < .001$, generalized eta squared = .170). The main effect of presentation method was also significant ($F(1, 130) = 46.448$, $p < .001$, generalized eta squared = .039), and the interaction was significant ($F(1, 130) = 4.924$, $p = .028$). Condition means are shown in Figure 5.

4.4 Discussion

In Study 1, the robot expressing approach-avoidance conflict was rated as more courageous than the robot in the no-conflict condition. This indicates that even when the behavior itself is the same, showing conflict between approach and avoidance motives before action makes the behavior easier to understand as involving fear or difficulty for the actor. This interpretation is consistent

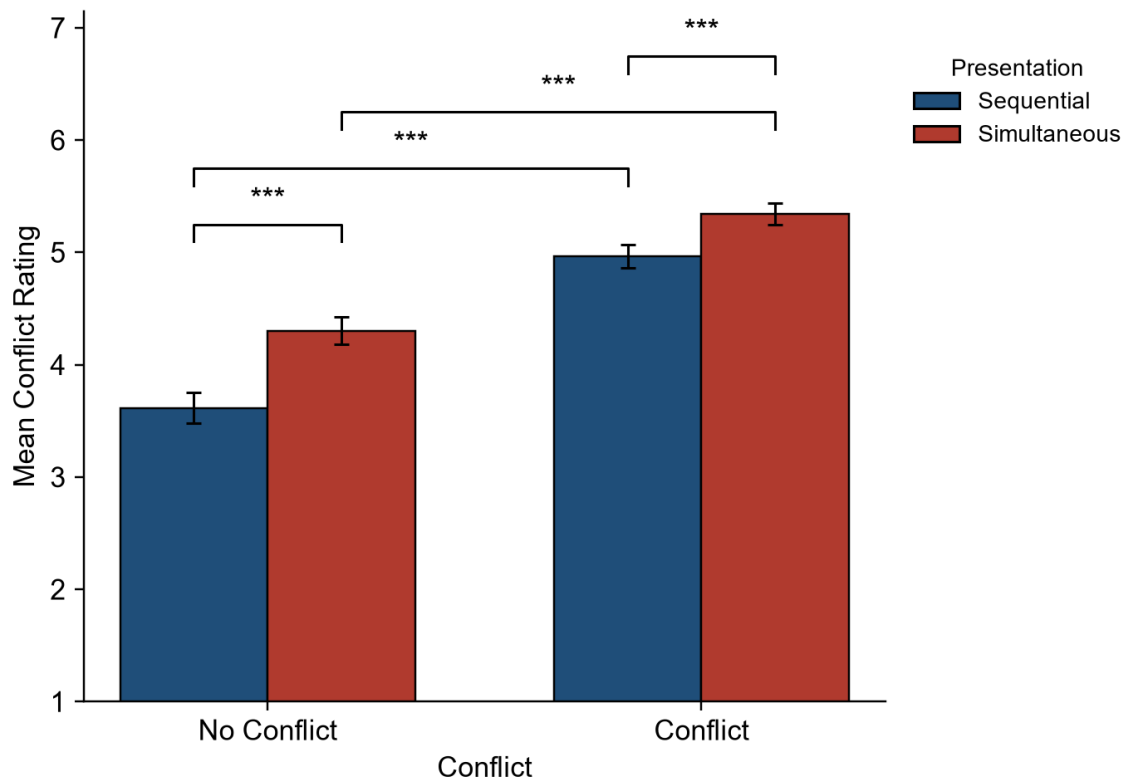


Figure 5. Condition means of conflict ratings in Study 1.

with the view that highly personal courageous actions involve fear or difficulty (Pury et al., 2007). Therefore, externalizing internal states before action related to personal courage may be effective for presenting the robot's behavior as courageous to observers.

Conflict ratings were also higher in the conflict condition than in the no-conflict condition. This indicates that the presentation of approach and avoidance motives through speech bubbles was perceived by observers as approach-avoidance conflict. Regarding presentation method, no difference was found between sequential and simultaneous presentation in courage ratings. In contrast, conflict ratings were higher for simultaneous presentation. Therefore, simultaneous presentation was judged appropriate for Study 2 not as a method that makes the robot appear more courageous, but as a method that conveys conflict more clearly.

Study 1 confirmed that a robot expressing approach-avoidance conflict is perceived as courageous and provided evidence supporting the validity of the presentation to be used in Study 2. Based on these findings, Study 2 used the simultaneous presentation selected in Study 1 to examine the main purpose of the present research: whether observing a robot perceived as courageous affects observers' self-evaluations of personal courage.

5 STUDY 2: DOES OBSERVING A ROBOT PERCEIVED AS COURAGEOUS INFLUENCE OBSERVERS' SELF-EVALUATIONS OF PERSONAL COURAGE?

5.1 Purpose

The purpose of Study 2 was to examine how observing the robot confirmed in Study 1 to be perceived as courageous influences observers' self-evaluations of personal courage. We also examined whether this influence differs depending on observers' pre-existing courage tendency.

In Study 2, the presence or absence of conflict and the presence or absence of action were manipulated separately. This was because courage is understood as moving toward a valued action in a situation involving fear or difficulty (Rachman, 1984; Woodard and Pury, 2007; Norton and Weiss, 2009). In Chowkase et al.'s model as well, courage is described as a sequential process involving situation perception, value evaluation, evaluation of action feasibility, and action decision (Chowkase et al., 2024). A condition showing only approach-avoidance conflict may reveal the observational effect of the internal state before action, whereas a condition showing only action may be limited to the observational effect of seeing the choice to move toward a valued action. Therefore, we separated the presence or absence of conflict and the presence or absence of action to distinguish among the effects of approach-avoidance conflict alone, action alone, and the combination of both.

In Study 2, we predicted that the effects of approach-avoidance conflict and action on observers' self-evaluations of personal courage would differ depending on pre-existing courage tendency. In particular, we predicted that, among participants with low pre-existing courage tendency, post-observation self-evaluation scores of personal courage would be highest in the conflict-with-action condition. The rationale was that the low-courage group, whose self-evaluations of courage were low, would be a group that experiences difficulty acting in conflict situations. Prior research has shown that observing a model who gradually copes while showing difficulty can increase self-efficacy and task performance (Schunk et al., 1987; Schunk and Hanson, 1989). It has also been shown that when a task is difficult, people are more likely to use others' judgments and behavior as cues, whereas people with high self-efficacy are relatively independent from social influence (Lucas et al., 2006). The conflict-with-action condition presents both the coexistence of avoidance and approach motives and the choice to move toward action. In other words, it presents a process that is close to the difficulty faced by the low-courage group while showing that this difficulty can be overcome. Therefore, we predicted that post-observation self-evaluation scores of personal courage would be highest in the low-courage group in this condition.

5.2 Method

5.2.1 Conditions

Study 2 used a three-factor mixed design with pre-existing courage tendency group as a between-participant factor and conflict and action as within-participant factors. Participants were classified into low- and high-courage groups based on their CM-J scores before stimulus presentation. Conflict was manipulated by whether both approach and avoidance motives were shown simultaneously or only one type of motive was shown. Action was manipulated by whether the robot produced the admonishing speech.

As in Study 1, the stimulus videos presented a scene in which the robot sees a person littering in a park. After showing situation recognition, the robot presented approach and avoidance motives

in the speech bubble according to the condition. In the conflict condition, approach and avoidance motives were presented simultaneously and alternately enlarged and reduced. In the no-conflict, action-present condition, only approach motives were presented. In the no-conflict, action-absent condition, only avoidance motives were presented.

In the action-present condition, after the motive presentation, the robot said, “Excuse me, this is not a place to throw away trash.” In the action-absent condition, after the motive presentation, the robot did not speak, and a state in which it did not perform the admonishing behavior was presented.

5.2.2 Measurement

In Study 2, we measured pre-stimulus personal courage tendency, post-stimulus self-evaluations of the observer’s own personal courage, and robot conflict ratings. All ratings were made on a 7-point scale. Personal courage was measured using the six items of the Japanese version of the Courage Measure (CM-J) (Shimotsukasa et al., 2023). The CM-J is positioned as a scale that measures the trait and individual differences of courage to act despite fear or anxiety (Shimotsukasa et al., 2023). In Study 2, the pre-stimulus CM-J score was used as an index of observers’ pre-existing courage tendency. In contrast, the post-stimulus CM-J score was treated not as a change in long-term courage trait itself but as the observer’s self-evaluation of personal courage immediately after stimulus presentation. In the analysis, the mean of the six items was used as the personal courage self-evaluation score.

Conflict ratings used the same four items as in Study 1. In the analysis, the mean of the four items was used as the conflict rating score. In addition, to confirm whether conflict ratings based on these four items corresponded to a direct judgment of conflict, the same direct item used in Study 1, “This robot appeared to be experiencing conflict,” was also obtained.

5.2.3 Procedure

Study 2 was conducted on Yahoo! Crowdsourcing. Participants first responded to the CM-J before stimulus presentation. They then viewed videos from the four conditions and, after each video, responded to self-evaluation items for their own personal courage and to robot conflict-rating items. The order of video presentation was fixed. This was done so that each participant could observe condition differences within the same context and more easily compare differences due to the presence or absence of approach-avoidance conflict and action within a single flow. The response data included video-content comprehension checks and attention-check items.

5.2.4 Participants

As in Study 1, exclusions were made based on video-content comprehension checks and attention-check items, resulting in a final analytic sample of 126 participants. Participants were 18 to 29 years old, with a mean age of 24.11 years ($SD = 3.62$). The sample included 55 men and 71 women. Based on pre-stimulus CM-J scores, participants with scores below 4 were classified as the low-courage group, and participants with scores of 4 or higher were classified as the high-courage group. The low-courage group included 69 participants, and the high-courage group included 57 participants.

5.2.5 Statistical Analysis

In Study 2, post-stimulus personal courage self-evaluation scores were analyzed as the main outcome, and conflict rating scores were analyzed as a manipulation check. For personal courage self-evaluation scores, a three-factor mixed analysis of variance was conducted with pre-existing courage tendency group based on pre-stimulus CM-J scores (below 4 vs. 4 or higher) as a between-participant factor and conflict (no conflict vs. conflict) and action (no action vs. action) as within-participant factors. Partial eta squared was reported as the effect size. When interactions were significant, simple effects were examined as needed. For simple-effects tests, normality of paired differences was examined using the Shapiro-Wilk test; paired t-tests were used when normality was satisfied, and Wilcoxon signed-rank tests were used when normality was not satisfied. Conflict rating scores were analyzed using the same factor structure to confirm the success of the conflict manipulation. Analyses were conducted using Python packages pandas, statsmodels, and SciPy.

5.3 Results

The three-way mixed analysis of variance on observers' personal courage self-evaluation scores revealed a significant main effect of pre-existing courage tendency group ($F(1, 124) = 124.884$, $p < .001$, partial eta squared = .502). By contrast, neither the main effect of conflict ($F(1, 124) = 0.067$, $p = .796$, partial eta squared < .001) nor the main effect of action ($F(1, 124) = 2.487$, $p = .117$, partial eta squared = .020) was significant. Among the two-way interactions, the interaction between pre-existing courage tendency group and conflict was significant ($F(1, 124) = 7.513$, $p = .007$, partial eta squared = .057). By contrast, neither the interaction between pre-existing courage tendency group and action ($F(1, 124) = 0.009$, $p = .925$, partial eta squared < .001) nor the interaction between conflict and action ($F(1, 124) = 2.613$, $p = .109$, partial eta squared = .021) was significant. The primary three-way interaction was also not significant ($F(1, 124) = 0.046$, $p = .831$, partial eta squared < .001). Thus, the hypothesis that personal courage self-evaluation scores would be particularly high in the conflict-with-action condition among the low-courage group was not supported.

Because the interaction between pre-existing courage tendency group and conflict was significant, we examined the simple effect of conflict within each group. Shapiro-Wilk tests of the paired difference scores indicated no significant departure from normality in the low-courage group ($W = 0.974$, $p = .152$) but indicated a significant departure in the high-courage group ($W = 0.928$, $p = .002$). Accordingly, a paired t-test was used for the low-courage group, whereas a Wilcoxon signed-rank test was used for the high-courage group. The simple-effects analyses showed a marginal trend toward higher personal courage self-evaluation scores in the conflict condition in the low-courage group (conflict $M = 3.104$, no conflict $M = 2.982$, $t(68) = 1.980$, $p = .052$, $d = .238$). In contrast, participants in the high-courage group reported significantly lower personal courage self-evaluation scores in the conflict condition (conflict $M = 4.757$, no conflict $M = 4.858$, $W = 404.0$, $p = .038$, $d = -.271$). These simple effects are shown in Figure 6.

As a manipulation check, conflict ratings were higher in the conflict condition than in the no-conflict condition. The main effect of conflict was significant ($F(1, 124) = 52.939$, $p < .001$, partial eta squared = .299). A supplemental check also showed that the conflict condition was rated higher than the no-conflict condition ($W = 1128.0$, $p < .001$, conflict $M = 5.151$, no conflict $M = 4.209$). Therefore, the conflict manipulation was successful in Study 2 as well.

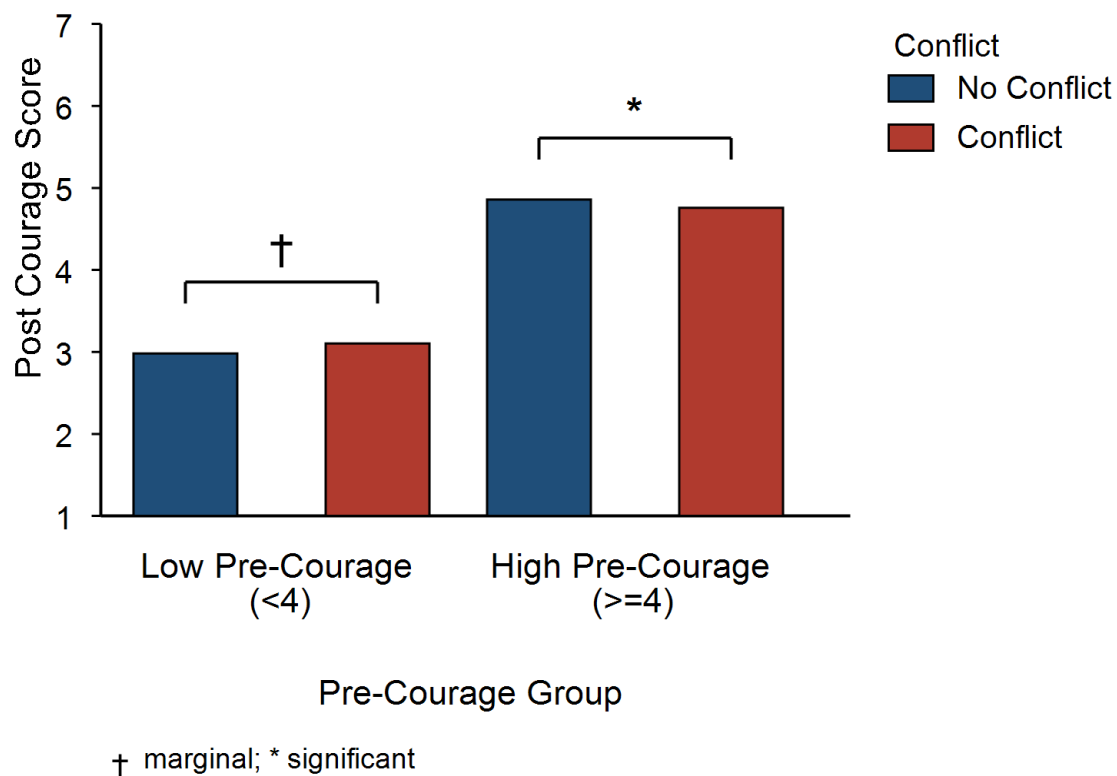


Figure 6. Simple effects of personal courage self-evaluations in Study 2. Means for the no-conflict and conflict conditions by pre-existing courage tendency group. Dagger indicates a marginal tendency; asterisk indicates a significant difference.

5.4 Discussion

In Study 2, the hypothesis that the low-courage group would show the highest personal courage self-evaluation scores in the conflict-with-action condition was not supported. This hypothesis treated the robot that expressed conflict and then admonished the litterer as analogous to a coping model that displays difficulty but ultimately performs the task. However, neither the main effect of action nor any interaction involving action was significant. The relative advantage of coping models has also not been consistently observed in prior research: Schunk and Hanson found no significant differences in self-efficacy or achievement between mastery- and coping-model conditions (Schunk and Hanson, 1985). Accordingly, the failure to detect a specific advantage for the conflict-with-action condition is compatible with evidence that coping models do not invariably outperform mastery models.

However, the interaction between pre-existing courage tendency group and approach-avoidance conflict was significant. This result indicates that presenting approach-avoidance conflict does not uniformly encourage observers but may operate in opposite directions depending on observers' pre-existing courage tendency. In the low-courage group, personal courage self-evaluation scores tended to be higher in the conflict condition. Previous studies have shown that observing a model who gradually copes while displaying difficulty can, in some circumstances, enhance self-efficacy and task performance more than observing a model who succeeds easily from the outset (Schunk et al.,

1987; Schunk and Hanson, 1989). Other work has shown that people are more likely to use others' judgments and behavior as cues when a task is difficult, whereas those with high self-efficacy are less susceptible to such influence (Lucas et al., 2006). The conflict condition in the present study showed a process in which the robot held an approach motive despite also having an avoidance motive. For the low-courage group, this may have served as a cue that a motive to move toward valued action can coexist with fear or hesitation. However, pre-existing courage tendency and self-efficacy are not identical constructs, and we did not measure whether participants perceived the robot as a model coping with difficulty. Moreover, because this result was only marginal, this interpretation should remain tentative.

By contrast, participants in the high-courage group reported lower personal courage self-evaluation scores in the conflict condition. Chowkase et al. theorized that courageous decisions involve weighing potential valued outcomes against the perceived risks associated with action (Chowkase et al., 2024). In an approach-avoidance task, Bach et al. experimentally showed that greater potential threat was accompanied by greater passive avoidance and behavioral inhibition (Bach et al., 2014). In the present conflict condition, the robot simultaneously expressed an approach motive indicating the value of admonishing the litterer and avoidance motives indicating risks such as being yelled at or ignored. These avoidance motives may therefore have been interpreted as information about the risks of acting, leading the high-courage group to consider not only the value of the action but also its associated risks and to evaluate their own courage more cautiously. However, neither of these prior studies directly showed that observing another agent's conflict lowers self-evaluation specifically among individuals with high pre-existing courage. They therefore provide only a possible general mechanism for interpreting the present result.

Although the main effect of conflict on personal courage self-evaluation was not significant, the interaction between pre-existing courage tendency group and conflict was significant, and the direction of the conflict contrast differed between groups. The absence of a main effect should therefore not be interpreted as evidence that conflict expression was unrelated to self-evaluation in both groups. Moreover, conflict ratings were higher in the conflict condition than in the no-conflict condition, indicating that participants perceived the intended conflict expression. Thus, the unsupported hypothesis cannot be attributed solely to a failed conflict manipulation. Overall, robot-expressed conflict did not uniformly raise observers' self-evaluations; instead, its association with self-evaluation may differ in direction according to observers' pre-existing courage tendency.

6 LIMITATIONS

This study has several limitations. First, although this study measured observers' self-evaluations of personal courage, it did not measure whether observers actually moved toward valued action. The CM-J is positioned as a scale that measures the trait and individual differences of courage (Shimotsukasa et al., 2023). Therefore, the post-stimulus CM-J score in this study should be interpreted not as a change in long-term courage trait but as observers' self-evaluations of personal courage immediately after stimulus presentation.

Second, this study did not directly measure the psychological process through which the presentation of approach-avoidance conflict operated in different directions for the low- and high-courage groups. In this manuscript, we suggested that for the low-courage group, the presentation may have served as a cue that "a motive to move toward valued action can coexist with fear

or hesitation,” whereas for the high-courage group, it may have made avoidance motives or the difficulty of action more salient. However, because we did not directly measure whether the robot was perceived as a model coping with difficulty, perceived similarity to the robot, self-efficacy, interpretation of avoidance motives, perceived action difficulty, or reception of risk information, these remain interpretations.

Third, this study used only video stimuli involving a robot and did not compare other presentation media such as a physical robot, human model, avatar, animation, or text presentation. Therefore, it is not possible to separate whether the present results derive from the use of a nonhuman agent as a robot, from presenting the stimulus as a video, or from explicitly presenting approach-avoidance conflict through a speech bubble.

Fourth, in Study 2, the order of video presentation was fixed. Therefore, the results of Study 2 may include not only the effects of approach-avoidance conflict and action but also the influence of videos viewed previously and order effects. Future studies should use a counterbalanced presentation order to examine whether the same effect is replicated.

Fifth, the stimulus scenario in this study was limited to admonishing littering in a park. This scenario has some validity as a stimulus for examining personal courage because valued action and interpersonal risk coexist. However, the present study dealt with a scene in which the robot directly admonished a stranger’s norm violation. Personal courage may include diverse situations such as speaking up in interpersonal contexts, seeking help, challenging learning tasks, social reintegration, and prosocial behavior (Pury et al., 2007; Howard et al., 2017; Vogel et al., 2007). Therefore, it remains unclear whether the present findings generalize to situations involving different types of risk or action value.

7 FUTURE WORK

Future research should proceed in two directions: basic and applied approaches. In the basic approach, it is necessary to examine the psychological process through which presenting approach-avoidance conflict affects observers’ self-evaluations of personal courage. Specifically, future studies should measure the extent to which observers perceive the robot as a self-relevant model or as a model coping with difficulty, perceived similarity to the robot, self-efficacy, perceived threat and difficulty, and the added value of action information, and clarify the conditions under which the interaction between pre-existing courage tendency and approach-avoidance conflict emerges. In addition, using the same scenario and internal-state expression, future work should compare a robot present in front of the observer, a robot in a video, a human model, avatars or animations, and text presentation, and examine the conditions under which a nonhuman robot functions as a social model as well as the roles played by the robot’s physical embodiment, co-presence, and movement.

In the applied approach, it is necessary to identify target populations and situations in which effects are expected and then examine them. In particular, for people with low pre-existing courage tendency, concrete support contexts may include challenging new learning tasks in educational settings, building interpersonal relationships in counseling or social reintegration contexts, and everyday prosocial behavior. In such contexts, future studies should measure not only post-observation self-evaluation but also actual action choices, such as speaking up, asking questions, seeking help, pointing out problems, and helping others, as well as sustained changes in these behaviors.

8 CONCLUSION

588 This study examined the possibility that a robot, as a nonhuman agent, can influence observers'
589 self-evaluations of personal courage by externalizing and presenting an internal state before
590 action, namely approach-avoidance conflict. Study 1 confirmed that a robot expressing approach-
591 avoidance conflict is perceived as courageous. Study 2 showed that observing such a robot does not
592 uniformly influence observers' self-evaluations of personal courage but may relate to them differently
593 depending on pre-existing courage tendency. These results do not demonstrate an effect unique to
594 robots, but they indicate the possibility of using robots as social models that present internal states
595 before action.

CONFLICT OF INTEREST STATEMENT

596 The authors declare that the research was conducted in the absence of any commercial or financial
597 relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

598 YS was responsible for study design, stimulus creation, data analysis, and manuscript writing. MB,
599 HT, and HI contributed to study design, interpretation of the results, and manuscript revision.

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DATA AVAILABILITY STATEMENT

602 The availability and sharing method of the data analyzed in this study will be added before
603 submission.

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Table 5. Study 2 conditions.

Study 2 condition	Internal-state content	Motive statements used	Presentation method	Final admonishing speech
No conflict, no action	Avoidance motives only	Avoidance motives: “I would hate it if they yelled at me after I warned them”; “Even if I warn them, they might ignore me and not take me seriously”	Avoidance motives presented and emphasized	Absent
No conflict, action	Approach motives only	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering”	Approach motives presented and emphasized	Present
Conflict, no action	Approach and avoidance motives	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering” / Avoidance motives: “I would hate it if they yelled at me after I warned them”; “Even if I warn them, they might ignore me and not take me seriously”	Simultaneous presentation with alternating emphasis	Absent
Conflict, action	Approach and avoidance motives	Approach motives: “If I warn them, the park might become cleaner”; “If I warn them, it might prompt that person to stop littering” / Avoidance motives: “I would hate it if they yelled at me after I warned them”; “Even if I warn them, they might ignore me and not take me seriously”	Simultaneous presentation with alternating emphasis	Present